MIL-M-6404A(USAF)
7 April 1965
Superseding
MIL-M-6404(USAF)
1 March 1951

MILITARY SPECIFICATION

METER, VOLUMETRIC, POSITIVE DISPLACEMENT, LIQUID, TYPE F-1, AIRCRAFT FUEL, 300 GALLONS PER MINUTE

1. SCOPE

1.1 This specification covers one type of aircraft fuel meter, designated Type F-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated - Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Materials, Water Vaporproof, Flexible
MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier Material
MIL-J-5624-	Jet Fuel, Grades JP-3, JP-4 and JP-5
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of

FSC 6680

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DRAWINGS

Air Force

No. 50D7936 Filter Assembly, Aircraft Fuel, 300 GPM, Type E-1

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Mechanical Engineers
American Petroleum Institute

ASME-API Code 1101 Code for Installation, Proving and Operation of Positive Displacement Meters in Liquid Hydrocarbon Service or Petroleum PD Meter Code

(Application for copies of the ASME-API Code should be addressed to the American Petroleum Institute, 50 West 50 Street, New York, New York.)

3. REQUIREMENTS

3.1 Component parts. The meter shall consist of a meter, meter register, and a totalizer.

3.2 Material.

3.2.1 Specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.2.1.1 Standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts list. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

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3.2.2 Protective treatment. When materials are used in the construction of the meter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3 Design and construction:

3.3.1 The meter shall be designed to conform to the physical limitations of Figure 1, and shall be capable of withstanding a static pressure of 125 psi. A meter having a single inlet opening shall be considered acceptable provided the design of the meter is such as to allow either right or left hand operation. It shall be permissible to accomplish this objectively by use of spacers, adapters, interchangeable end plates, etc., with adequate provisions for maintaining proper direction of counter rotation. All such appurtenances shall be furnished as installed equipment on the meter with proper instructions for their use permanently attached to the meter.

3.3.2 The meter shall be designed so that ease of production, easy and rapid installation, and ease of general maintenance and repair will be assured. The design shall be such that overhaul and repair can be accomplished with common tools and shall not require special tools or fixtures.

3.3.3 The meter shall be designed and constructed to meet the performance requirements of this specification and withstand the stresses, impacts, vibrations, and other conditions incident to shipping, storage, and service usage.

3.3.4 Meter assembly. The meter assembly shall be designed to meet the performance requirements when operated in either direction. It shall be equipped with a 3 inch flanged-type inlet and outlet.

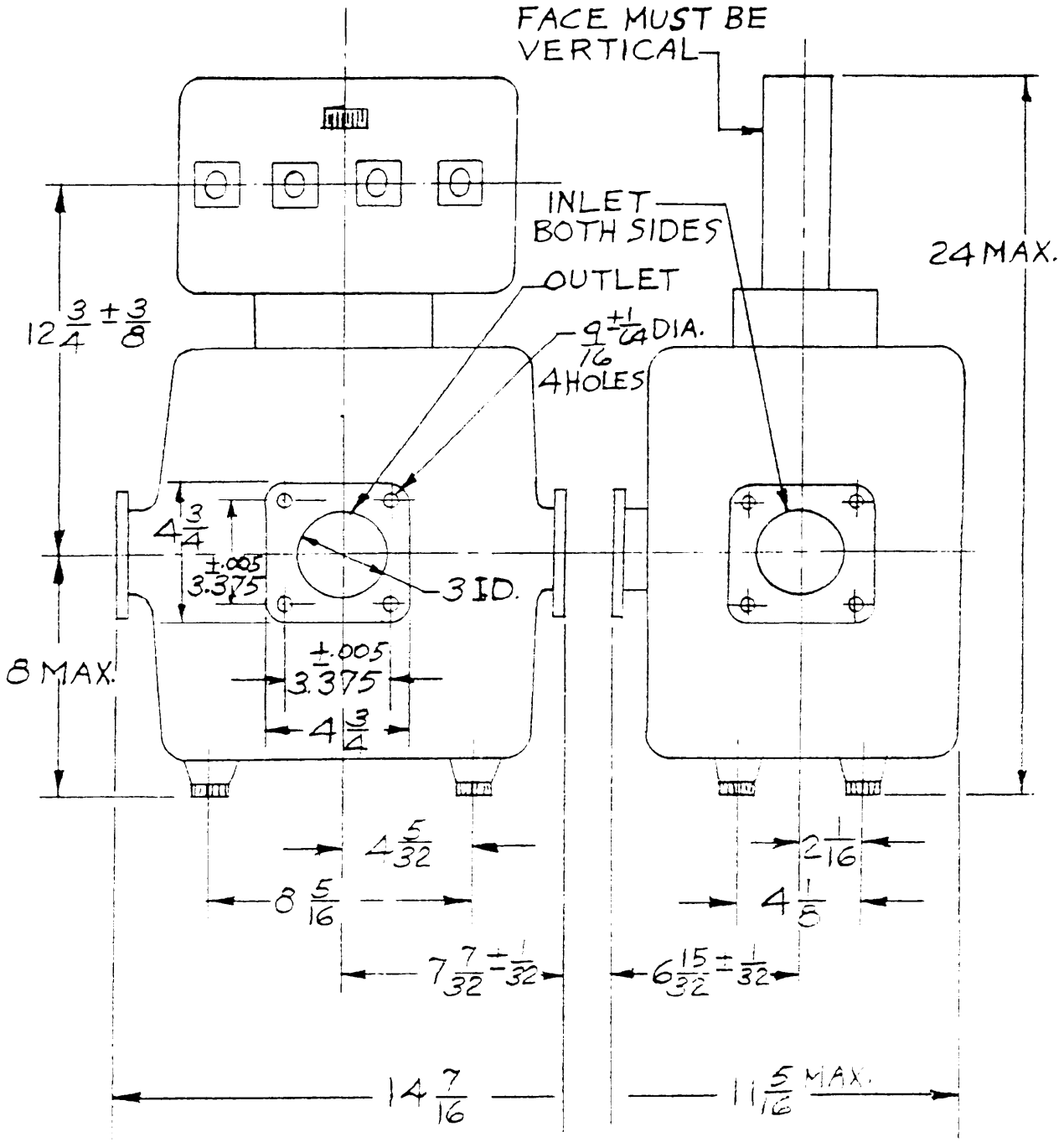
3.3.4.1 Drain. A drain shall be provided at the lowest point of the meter housing. This drain shall be leak proof when closed.

3.3.4.2 Direction of flow. The meter shall be capable of operation in either direction and the meter register shall indicate additive or subtractive quantities, depending upon which direction the meter assembly is operating. There shall be no shifting of gears to attain this reversal.

3.3.4.3 Face. The counter face shall be of a transparent colorless plastic. No glass shall be used.

3.3.4.4 Measuring chamber. There shall be no metal to metal contact between any parts which are in motion relative to each other, other than the rotor bearings or counter driving gear.

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Dimensions in inches unless otherwise specified.
Tolerance: Fraction 1/16

FIGURE 1 METER ASSEMBLY

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3.3.4.5 Counter indexing. The counter assembly shall be capable of being indexed throughout a full revolution in increments of 30 degrees, or less, to permit positioning of the counter in any direction with relation to the meter body.

3.3.4.6 Accuracy adjustment. The meter assembly shall be equipped with a positive coupled accuracy adjustment mechanism which shall not change over the meter life except by manual adjustment. The accuracy adjustment mechanism shall be designed and calibrated for equal manual adjustment in positive increments within the limits of 1/2 gallon or less per 1000 gallons. The design of the accuracy adjustment mechanism shall permit ready adjustment without searching or partial disassembly of the mechanism, except the removal of a cover plate. Provision shall be provided for sealing the adjustment mechanism or cover plate with a lead seal, 1/8 inch diameter and 1 1/2 inches of wire. The accuracy adjustment shall have a minimum range of 5 percent.

3.3.4.7 Meter register. The meter register shall contain a set of 4 direct reading digits, coated with fluorescent paint. Each digit shall have a minimum height of 1/2 inch which shall be capable of being reset after flow in either direction, and capable of recording a flow of 9999 gallons of fuel. The register shall also be provided with a totalizer containing 7 digits. A positive counter drive with no friction devices shall be utilized to drive this register.

3.3.4.8 Meters shall be interchangeable to the extent that complete meters can be interchanged on a fuel servicing vehicle without checking the calibration of the meter by use of a power tank or master meter.

3.4 Performance.

3.4.1 Flow pressure drop. The maximum pressure drop between the meter inlet and outlet shall not exceed those shown on Figure 2.

3.4.2 Foreign substance. The performance of the meter shall not be critically affected by the foreign substance passed through a Type F-1 filter conforming to USAF Drawing No. 50D7936.

3.4.3 Meter life. The meter assembly shall be designed for a life expectancy which shall permit metering of 30,000,000 gallons of fuel at a flow rate of 300 gpm and in accordance with the overall requirements of this specification.

3.4.4 Temperature. The meter assembly shall be capable of meeting all performance requirements of this specification at ambient temperatures throughout the range of -54° to +54° C. In addition, the meter assembly shall not be adversely affected by extended storage at ambient temperatures of -65° and +71° C.

3.4.5 Fungus resistance. The meter assembly shall be constructed of materials which are completely resistant to the deteriorating effects of fungus occurring in tropical climates or they shall be treated in a manner which will adequately prevent fungus attack.

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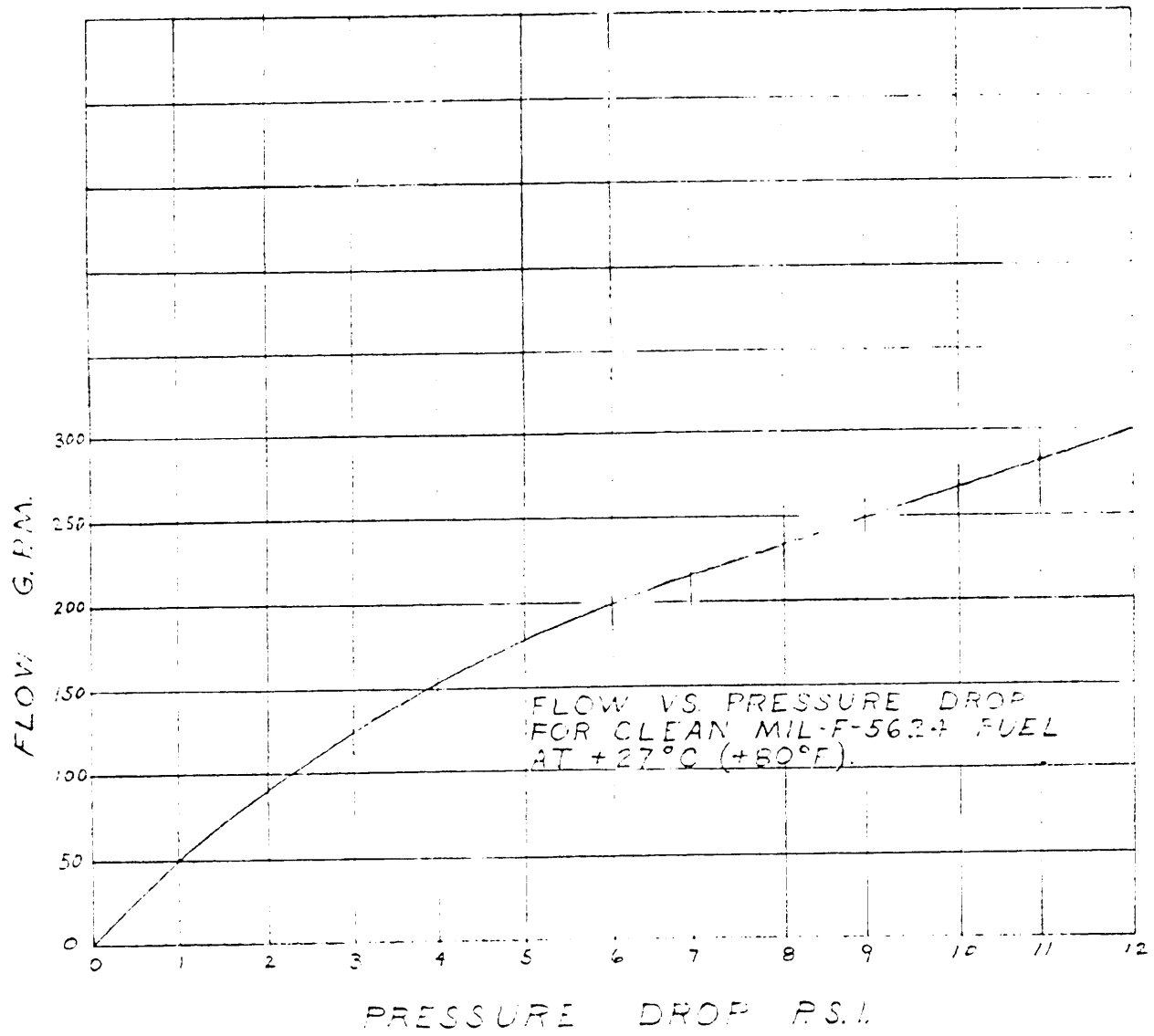


FIGURE 2 FLOW CHART

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3.4.6 Salt spray. The meter shall be capable of satisfactory operation after being subjected to salt laden atmosphere and shall not be adversely affected by sea water in the fuel.

3.4.7 Sand and dust. The meter and counter assembly shall be completely sealed against the deleterious effects of sand and dust. The meter assembly shall be completely operable after being subjected to the sand and dust test.

3.4.8 Error of the meter assembly. The maximum error of the meter assembly for flow in either direction shall not exceed 1.1 of 1 percent for any one predetermined flow rate and accuracy setting. The maximum error of the meter assembly shall not exceed 1.3 of 1 percent for any one given accuracy adjustment, at any flow rate over a range from 15 to 300 gpm.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-I-Tight.

3.6 Weight. The weight of the meter assembly complete with adjustment mechanism, counter, and totalizer shall not exceed 100 pounds.

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.8 Workmanship.

3.8.1 General. The meter assembly, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, welding and brazing, painting, machine screw assemblies and freedom of parts from burrs and sharp edges.

3.8.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the meter assembly, they shall be held or limited accordingly.

3.8.3 Cleaning. The meter assembly shall be thoroughly cleaned of loose metal chips and other foreign substance. Burrs and sharp edges shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of meters shall be classified as follows.

4.2.1 Preproduction tests. Preproduction tests are those tests accomplished on a sample representative of the production meter to determine that the production meter meets the requirements of this specification.

4.2.2 Inspection tests. Inspection tests are those tests accomplished on meters submitted for acceptance under contract.

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing prescribed herein shall be, insofar as practicable, of laboratory precision type and shall be calibrated at intervals properly spaced to assure laboratory accuracy.

4.3.2 Fuel. The fuel used during the testing shall conform to MIL-F-5624.

4.4 Preproduction testing.

4.4.1 Preproduction test sample. The preproduction test sample shall consist of one model representative of the production equipment.

4.4.2 Samples submitted to the procuring agency for test shall be accompanied by the applicable bill of material and manufacturer's drawings, ample operating instructions, and adequate spare gaskets, locking device, seals, etc., to provide replacements which may be necessary for the complete disassemblies and reassemblies required during the testing outlined herein.

4.4.3 Part failure. If during the preproduction test of a meter assembly, a part fails, that part shall be replaced with a redesigned part or one of a different material, unless the procuring agency authorizes the installation of a new part of the original design and material for one which failed due to faulty material or workmanship. The preproduction test shall be considered completed when every major part in the meter assembly has passed the complete test. At the discretion of the procuring agency, redesigning and retesting may be required of any part which fails or indicates weakness after completing its preproduction test, but which is retained in the meter assembly to complete testing of other parts.

4.4.4 Data.

4.4.4.1 Readings. During the testing prescribed herein, at least the following data shall be recorded at intervals not greater than 30 minutes, except where shorter periods are a test requirement:

Time of day
Ambient temperature
Fuel flow rate
Fuel pressure at inlet to meter

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Fuel pressure at outlet of meter
Pressure differential
Fuel temperature at inlet to meter
Fuel specific gravity

4.4.4.2 Records. Notes shall be placed on the log sheets of all incidents of the test, such as adjustments made, servicing performed, leaks, and any other irregular functioning of the metering assembly and corrective measures taken.

4.4.4.3 Preproduction test report. If the contractor completes the preproduction tests, he shall prepare a preproduction test report in accordance with MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.4.5 The preproduction test shall consist of the following tests in the order listed, and those specified under inspection tests subject to the test conditions specified herein.

4.4.5.1 Flow. Clean fuel filtered through a Type I-1 filter shall be pumped through the meter assembly at flow rated from 50 to 300 gpm in increments of 50 gpm. The pressure drop between the inlet and outlet of the meter assembly will be measured after the flow rate has been stabilized. The maximum pressure drop between the inlet and outlet of the metering assembly during the test shall not exceed the pressure shown on Figure 2 for each of the respective flow rates.

4.4.5.2 Static pressure. The meter assembly shall be subjected to a hydrostatic pressure of 125 psi for a period of 30 minutes. The meter assembly shall exhibit no leakage, rupture, or evidence of failure in any way.

4.4.5.3 Calibration. Fuel shall be pumped through the meter into a prover tank at a flow rate of 300 gpm. The calibration of the meter shall be determined and the maximum error shall not exceed 0.1 percent.

4.4.5.3.1 The maximum error at any flow rate over this range shall not exceed 0.3 percent. Fuel shall be pumped through the meter into a prover tank at flow rates from 15 to 50 gpm and in increments of 50 gpm thereafter to 300 gpm.

4.4.5.3.2 The calibration test shall be in accordance with the applicable sections of ASME-API Code 1101.

4.4.5.4 High temperature. The meter assembly shall be subjected to high temperature tests, Procedure I of MIL-STD-810. Following the 50 hour exposure at +70° C, the meter assembly shall be removed from the chamber and visually examined for deterioration during the test. The meter assembly shall exhibit no deleterious effects of the exposure and shall subsequently complete the remainder of the tests in a satisfactory manner.

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4.4.5.5 Low temperature. The meter assembly shall be subjected to low temperature tests, Procedure I of MIL-STD-810. Following this test the meter shall exhibit no evidence of failure and shall subsequently complete the remaining tests in a satisfactory manner.

4.4.5.5.1 The meter assembly and a supply of fuel to be used in testing shall be placed in a low temperature test chamber and exposed to an ambient temperature of -54° C for a period of 48 hours. The meter assembly shall then be subjected to the following tests.

4.4.5.5.1.1 Low temperature flow. Fuel at a temperature of -54° C shall be pumped through the meter assembly for a period of 20 minutes at a flow rate of 300 gpm. The pressure differential between the inlet and outlet of the meter assembly shall not exceed 12 psi during the test.

4.4.5.5.1.2 Low temperature static pressure. The meter assembly shall be subjected to a hydrostatic pressure of 125 psi for a period of 30 minutes. The meter assembly shall exhibit no leakage, rupture, or evidence of failure in any way.

4.4.5.6 Humidity. The meter assembly shall be subjected to Humidity tests, Procedure I of MIL-STD-810. Following the test the meter assembly shall be visually examined for deterioration, evidence of swelling, separation of components and related types of failure.

4.4.5.7 Fungus. The meter assembly shall be subjected to Fungus tests, Procedure I of MIL-STD-810. The meter assembly shall resist the effects of the fungi used and with superficial cleaning shall complete the remaining tests in a satisfactory manner.

4.4.5.8 Salt fog. The meter assembly shall be exposed to a salt fog test in accordance with salt fog test Procedure of MIL-STD-810 for a period of 50 hours. The meter assembly shall resist the effects of the tests and exhibit no failures or malfunctioning. During this test the inlet and outlet of the meter shall be plugged to prevent internal contamination.

4.4.5.9 Sand and dust. The meter assembly shall be exposed to sand and dust tests in accordance with sand and dust tests, Procedure I of MIL-STD-810. The meter assembly shall resist the effects of the test and exhibit no failures or malfunctioning. During this test the inlet and outlet of the meter shall be plugged to prevent internal contamination.

4.4.5.10 Life test.

4.4.5.10.1 Meter. Fuel shall be pumped through the meter at a flow rate of 300 gpm for a total of 500 hours. There shall be no sign of wear, failure, or other defects. Upon completion at the endurance test and without intervening disassembly, the meter shall satisfactorily complete tests in accordance with 4.4.5.3 and 4.4.5.3.1. It is permitted to readjust the accuracy mechanism of the meter register to allow the meter to meet the requirements of the calibration tests.

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4.4.5.11 Disassembly inspection. Upon completion of the testing prescribed above, the meter assembly shall be disassembled and its components shall be examined. The meter shall exhibit no indication of wear.

4.5 Inspection tests.

4.5.1 Individual tests. Each meter shall be subjected to the following tests.

4.5.1.1 Examination of product. Each meter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.5.1.2 Flow. Each meter assembly shall be subjected to a fuel flow calibration test, using a prover tank which will insure that the complete assembled unit is free from physical defects and calibrated for a 300 gpm servicing rate. The specific testing method used by the contractor shall be inspected and approved by the procuring agency prior to its use by the contractor.

5. PREPARATION FOR DELIVERY

5.1 Immediately prior to packaging, all meter assemblies shall be inspected to insure that no water, fuel, dangerous vapors, or dirt contamination are present within.

5.1.1 Preservation and Packaging.

5.1.2 Level A and B. Unless otherwise specified by the procuring agency, the meter assembly shall be preserved and packaged in accordance with MIL-P-116 Method IIA. Container utilized shall conform to PPP-B-601 or PPP-B-621.

5.1.3 Level C. Preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to first domestic receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A and B. Meters prepared as indicated in para 5.1.2 above will require no additional packing for shipment.

5.2.2 Level C. Meters prepared as indicated in para 5.1.3 above shall be packed in a manner that will afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. This level is general conforms to applicable carriers rules and regulations and may be suppliers commercial practice when such meets the requirements of this level.

5.3 Marking and shipment. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows: Meter, Aircraft Fuel, 300GPM, Type F-1.

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5.3.1 Reinspection date marking. Reinspection marking will be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The type F-1 meter covered by this specification is intended for use in the filed to indicate the gallons per minute of jet fuel pumped into and from certain refueling and defueling vehicles.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2.1 Preproduction tests. One meter will be required as a preproduction sample to be subjected to the preproduction tests to determine compliance with the requirements of this specification. The point of inspection for these tests will be specified.

Custodian:

Air Force - 67

Reviewer:

Air Force - 67

Preparing Activity:

Air Force - 67

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines at reverse side, staple in corner and send to preparing activity (see ind. also on reverse hereof).

SPECIFICATION
MIL - M - 6404A (USAF)

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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